

# IP-SAKTI SAHAYAK

## Family 7 — Phyllanthaceae / Amla

**Objective:** Build a TKDL-only family cluster around Amla so IP-SAKTI can identify related TKDL-listed plants and rank them as potential alternatives for further research when a formulation or ingredient has a relevant traditional-knowledge / prior-art signal.

**Strict TKDL rule:** This family is particularly strong for dataset expansion because the official TKDL plant-name index contains multiple *Phyllanthus* entries. The candidate pool below contains only plants explicitly appearing in that official TKDL index. No outside species have been added merely to reach a target count.

**Taxonomy note:** TKDL records may use historical nomenclature/family placement. For example, a TKDL examination record describes *Phyllanthus emblica* under the Euphorbiaceae family. IP-SAKTI should preserve the family/taxonomic wording of the underlying TKDL source in the evidence field and keep standardized taxonomy as a separate field.

### 1. Anchor / main plant

**Amla — *Phyllanthus emblica* Linn.** (syn. *Embolica officinalis* Gaertn.)

TKDL Ayurvedic name: Amalak. Common/TKDL names include Amla and Nelli. The official TKDL index lists the plant directly.

### 2. Related TKDL species — 6 candidates

| TKDL plant                               | Relationship to anchor                   | Similarity*      | Evidence / model use                                                           |
|------------------------------------------|------------------------------------------|------------------|--------------------------------------------------------------------------------|
| <i>Phyllanthus amarus</i> Schum. & Th.   | Same genus; very close botanical cluster | VERY HIGH — 0.92 | TKDL index + public examination evidence                                       |
| <i>Phyllanthus fraternus</i> Webster     | Same genus; very close botanical cluster | VERY HIGH — 0.92 | TKDL index + public examination evidence                                       |
| <i>Phyllanthus maderaspatensis</i> Linn. | Same genus                               | HIGH — 0.84      | TKDL index; specific formulation match to verify                               |
| <i>Phyllanthus polyphuilus</i> Willd.    | Same genus                               | HIGH — 0.84      | TKDL index; specific formulation match to verify                               |
| <i>Phyllanthus tenellus</i>              | Same genus                               | HIGH — 0.84      | TKDL index; specific formulation match to verify                               |
| <i>Phyllanthus urinaria</i> Linn.        | Same genus                               | VERY HIGH — 0.90 | TKDL index; same Bhumi-malak terminology; specific formulation match to verify |

\*Prototype ranking score only. It is not a scientific equivalence score and must not be presented as legal clearance.

### 3. Official TKDL plant-index verification

Official TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official TKDL index lists: *Phyllanthus acidus*, *Phyllanthus amarus*, *Phyllanthus emblica*, *Phyllanthus fraternus*, *Phyllanthus maderaspatensis*, *Phyllanthus polyphuilus*, *Phyllanthus tenellus* and *Phyllanthus urinaria*. For this family dataset, six related candidates are paired with *Phyllanthus emblica* as the anchor. [cite0search0turn1search11](#)

### 4. TKDL evidence — Amla / *Phyllanthus emblica*

**EP1863507 — EPO:** TKDL third-party observations brought prior-art references showing *Phyllanthus emblica* (Amla) used with other ingredients for treatment of acne and as a complexion-promoting agent. The EPO examiner took cognizance of the TKDL references and stated that the disclosure was relevant to inventive-step assessment under Article 56 EPC. Examination was reported as in progress.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/EP1863507.asp>

**2221/DEL/2008:** The official TKDL examination outcome describes a herbal formulation for prevention and treatment of diabetes and associated complications containing fruits of *Embolica officinalis* Gaertn. (synonym: *Phyllanthus emblica* L.) at 5–8% w/w, alongside other ingredients. This is useful as a formulation-level record for Amla.

Official TKDL outcome: <https://www.tkdil.res.in/tkdil/Outcomes/2221-DEL-2008.pdf>

**212/DEL/2006:** TKDL pre-grant opposition cited *Phyllanthus emblica* (Amla) among traditional-use references for tumour/cancer-related conditions. The examination report records objections under Section 3(p) based on TKDL search results and states that the application was refused on the Controller's decision.

Official TKDL record: [https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/212\\_DEL\\_2006.asp](https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/212_DEL_2006.asp)

**2381/CHE/2010:** The official examination record states that *Phyllanthus emblica*, among other ingredients, had been used alone and in combination for anti-obesity treatment and that the claimed combination lacked inventive step in view of cited TKDL documents.

Official TKDL record: [https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/2381\\_CHE\\_2010.asp](https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/2381_CHE_2010.asp)

### 5. Related-species evidence

***Phyllanthus amarus*:** The official TKDL plant index lists it as a separate entry. A public TKDL-linked examination record at IP Australia (2009222618) cites *Phyllanthus amarus* (Tamalaki) and *Phyllanthus niruri* (Bhumyamalaki) for traditional uses including viral diseases/infections, obesity, strength promotion and removing tiredness. This provides formulation/use-context evidence, but the record should be kept separate from the identity of *Phyllanthus emblica*.

Official TKDL record: <https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/2009222618.asp>

**Phyllanthus fraternus:** The official TKDL index lists it separately and gives the Ayurvedic terminology T■malak■ / Bh■my■malak■. Because the same terminology is also used for Phyllanthus amarus in the TKDL index, the model must preserve the botanical name and not merge the two records solely on the vernacular name.

**Phyllanthus urinaria:** The official TKDL index lists it separately as Bh■my■malak■ and gives the common name chamber bitter. It is therefore a strong same-genus retrieval candidate, but the model should not assume therapeutic interchangeability with Amla without formulation/use evidence.

**Phyllanthus maderaspatensis / P. polyphuilus / P. tenellus:** All three are explicitly present in the official TKDL plant-name index. For this first-pass dataset they remain verified candidate nodes, while their specific formulation/use records should be linked before they receive a high-confidence recommendation.

## 6. Similarity logic for IP-SAKTI

### Recommended hierarchy:

1. Same genus + specific TKDL formulation/use overlap → highest priority.
2. Same genus + strong traditional-use overlap → high priority.
3. Same genus + only plant-index evidence → provisional candidate.
4. Vernacular-name overlap without botanical confirmation → do not merge records.
5. Any identity conflict → human verification required.

**Example:** If a user's Amla formulation triggers a prior-art warning, IP-SAKTI retrieves the Phyllanthus family cluster and may surface *P. amarus*, *P. fraternus* or *P. urinaria* as candidates. The recommendation should explain that they are same-genus candidates and then show the specific TKDL evidence supporting or weakening similarity.

## 7. Model-ready dataset fields

family | anchor\_plant | candidate\_plant | botanical\_name | TKDL\_name/synonyms | plant\_part | traditional\_use | formulation | formulation\_role | source\_text | TKDL\_case\_id | official\_TKDL\_url | similarity\_score | similarity\_reasons | patent\_IP\_flag | evidence\_level | verification\_status

## 8. AI / legal guardrails

- Same family/genus does not mean interchangeable.
- Same family/genus does not establish freedom from patent infringement.
- Never output “no legal barriers” without separate claim-level patent/legal analysis.
- Distinguish traditional-knowledge evidence from novelty, inventive step and Section 3(p).
- Preserve botanical identity even when TKDL vernacular names overlap.
- Show the exact official TKDL record behind each recommendation.
- Recommended UI wording: “**Potential alternative candidate for further research — botanical, formulation, efficacy and patent/legal verification required.**”

## 9. Verification status

This family provides a strong TKDL-only seed cluster: one anchor plus six related Phyllanthus entries. The next data-enrichment pass should attach exact TKDL formulation/use records to each candidate before the similarity engine treats it as a high-confidence alternative.